

Lab 1: Configuring IAM Roles and Policies

Duration: 30 minutes · **Region:** us-east-2 (Ohio) · **Prefix everything with your username:** userN-

Objectives

- Create an IAM group and user with a managed policy
 - See least privilege in action by signing in as that user
 - Create an IAM role for EC2 (used in Labs 3 and 5)
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Exercise 1.1: Sign In (3 min)

1. Open the sign-in URL from your instructor and sign in with your IAM user name and password.
2. Top-right **Region selector** → **US East (Ohio) us-east-2**.
3. Search for **IAM** and open it. Note the Region now shows **Global** — IAM is not Regional.

Exercise 1.2: Create a Group (4 min)

1. **User groups** → **Create group**.
2. Name: userN-developers .
3. Under permissions policies, check **AmazonS3ReadOnlyAccess** .
4. **Create user group**.

Exercise 1.3: Create a User (5 min)

1. **Users** → **Create user**. Name: userN-dev .
2. Check **Provide user access to the AWS Management Console** → *I want to create an IAM user*.
3. **Custom password**: pick one you'll remember. **Uncheck** *must create a new password at next sign-in*.
Next.
4. **Add user to group** → check userN-developers → **Next** → **Create user**.
5. Copy the **Console sign-in URL** shown on the confirmation page.

Exercise 1.4: Test Least Privilege (10 min)

1. Open a **private/incognito window** and go to the sign-in URL. Sign in as userN-dev . Set Region to us-east-2.
2. Open **S3**. You can see the bucket list — the policy allows listing.
3. Choose **Create bucket**, type any name, and click **Create bucket** at the bottom. Read the error:
s3:CreateBucket is not allowed.
4. Open **EC2** → **Instances**. You are denied — the policy says nothing about EC2, so it is implicitly denied.

5. Sign out and close the private window.

IAM starts from "deny everything." `userN-dev` can do exactly what `AmazonS3ReadOnlyAccess` allows and nothing else.

Exercise 1.5: Create a Role for EC2 (5 min)

1. Back in your main window: **IAM** → **Roles** → **Create role**.
2. **Trusted entity**: AWS service. **Use case**: EC2. **Next**.
3. Check `AmazonS3ReadOnlyAccess` . **Next**.
4. **Role name**: `userN-ec2-role` → **Create role**.

In Lab 3 you attach this role to a server; in Lab 5 the server reads S3 with it — no access keys needed.

Keep for later labs

`userN-developers` , `userN-dev` , `userN-ec2-role` — all deleted in Lab 8.

Knowledge Check

1. Why could `userN-dev` list buckets but not create one?
2. What is the difference between a user and a role?